

Ultrasound-guided shoulder

NOVA-E1 1 mL + NOVA-M1 1 mL · physician treatment protocol

A standardized, physician-directed framework for chronic shoulder pain, informed by a prospective 3-patient case series that reported no treatment-related adverse events and improvement in pain, mobility, stiffness, and daily function through 12 weeks after a single administration.

Indications

- Chronic shoulder pain > 6 months
- Rotator cuff tendinopathy or degenerative cuff pathology
- Mild-to-moderate glenohumeral osteoarthritis
- Chronic inflammatory shoulder pain
- Subacromial bursitis refractory to conservative treatment
- Failed PT, activity modification, medication, or prior injections

Patient preparation

1. Verify indication; review imaging when available.
2. Obtain informed consent.
3. Record baseline VAS pain score and shoulder function.
4. Position seated or lateral decubitus.
5. Prepare skin with chlorhexidine; establish sterile field.
6. Use ultrasound guidance throughout.

POST-PROCEDURE CARE

- Observe 15 minutes following injection
- Mild soreness may occur for 24–72 h
- Avoid NSAIDs and alcohol for 72 h when appropriate
- Restrict heavy overhead lifting 2–3 days; encourage gentle ROM

Contraindications

- Active local or systemic infection
- Pregnancy or breastfeeding
- Prior shoulder arthroplasty in the target joint
- Corticosteroid injection within 4 weeks
- Systemic corticosteroid use within 30 days
- Uncontrolled bleeding disorder or significant anticoagulation risk

Injection & distribution

TOTAL VOLUME

2 mL

Glenohumeral joint ≈ 1 mL; remaining volume periarticular to areas of pathology — cuff insertion, subacromial bursa, posterior capsule, or biceps tendon sheath per ultrasound findings.

Equipment: high-frequency linear transducer, sterile cover and gel, chlorhexidine, sterile gloves and drapes, 22–25 G needle.

FOLLOW-UP & OUTCOME MEASURES

Week 2

Safety review

Week 6

Functional progress

Week 12

Primary outcome

VAS PAIN

ROM

OVERHEAD TOLERANCE

ADLs

SLEEP / NIGHT PAIN

SATISFACTION

ADVERSE EVENTS

REGULATORY

Intended for physician-directed clinical and investigational use. Providers are responsible for compliance with applicable federal, state, facility, IRB, and professional practice requirements.

VISIT NOTE — SOAP
Shoulder · IA / peritendinous

CHART DOCUMENTATION · EVALUATION VISIT

FORM CN-SN-SHL · REV 2026.1

PATIENT NAME	DATE OF BIRTH	PATIENT ID / MRN	DATE OF VISIT
EVALUATING PHYSICIAN	INITIAL · FOLLOW-UP VISIT TYPE — CIRCLE	REFERRING PROVIDER, IF ANY	TIME IN / OUT

S SUBJECTIVE

Shoulder pain —
CHIEF COMPLAINT

ONSET / DURATION

SEVERITY — VAS 0-10

QUALITY / RADIATION

AGGRAVATING FACTORS

RELIEVING FACTORS

PT · NSAIDs · corticosteroid inj · activity modification · prior surgery
PRIOR TREATMENT — CIRCLE ALL TRIED · ADD DATES

overhead reach · dressing · sleep on side · lifting · work / sport
FUNCTIONAL IMPACT

fever · chills · weight loss · neuro changes — note positives
PERTINENT ROS

MEDICATIONS — INCL. ANTICOAGULANTS, IMMUNOSUPPRESSANTS

ALLERGIES · RELEVANT PMH — CANCER, INFECTION, PREGNANCY

O OBJECTIVE

BP

HR

TEMP

WEIGHT

GENERAL APPEARANCE

Inspection — atrophy, symmetry

Palpation — AC joint, biceps groove, subacromial

ROM — forward flexion / abduction / external rotation, degrees

Strength — supraspinatus, external rotation, lift-off

Provocative — Neer, Hawkins, drop arm

IMAGING / STUDIES REVIEWED — MODALITY, DATE, FINDINGS

BASELINE — VAS /10

BASELINE — DASH

BASELINE — ASES

A ASSESSMENT

Rotator cuff tendinopathy · partial tear · glenohumeral OA
WORKING DIAGNOSIS — CONFIRM OR AMEND

ADDITIONAL DIAGNOSES / CODES

☐ Appropriate candidate — indications met, contraindications absent

☐ Alternatives discussed, including conservative care and doing nothing

☐ Non-FDA-approved status explained; no outcome promised

☐ Patient questions answered; understanding confirmed

P PLAN

Proposed treatment. Ultrasound-guided injection, affected shoulder: NOVA-E1 + NOVA-M1, 2-4 cc total, intra-articular or peritendinous target per plan, 22-25 G. Observation 10-15 min.

☐ Informed consent (CN-PC) reviewed and signed

☐ Procedure performed today — see CN-OP-SHL · or scheduled (date right)

☐ Fees, non-coverage by insurance, and refund policy reviewed

☐ Follow-up: assessment at 4-6 wk; repeat consideration 6-12 mo

PHYSICIAN SIGNATURE	PRINTED NAME & LICENSE NUMBER	PROCEDURE DATE, IF SCHEDULED
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PATIENT NAME

DATE OF BIRTH

PATIENT ID / MRN

DATE OF PROCEDURE

PERFORMING PHYSICIAN

Rotator cuff tendinopathy · partial tear · glenohume...

PRE-PROCEDURE DIAGNOSIS — CONFIRM OR AMEND

START TIME

ASSISTING STAFF

PRE-PROCEDURE CHECKLIST

Check each box before the first draw

- ☐ Informed consent (CN-PC) signed and on file; SOAP note (CN-SN-SHL) complete
- ☐ Screening reviewed — no contraindications present
- ☐ Time-out performed — correct patient, site, product, route
- ☐ Five checks read aloud with second staff: ID · product & lot · expiry · volume & route · consent
- ☐ Baseline outcome measures recorded: VAS pain · DASH · ASES
- ☐ Site marked and prepped — chlorhexidine / povidone-iodine (circle); chain-of-custody log initiated

BP · HR · TEMP

agent · concentration · volume — write NONE i...

BP · HR · TEMP

PRE-PROCEDURE VITALS

LOCAL ANESTHETIC

POST-OBSERVATION VITALS

PRODUCTS ADMINISTERED

US-guided; joint and peritendinous targets per plan

PRODUCT & TARGET	PLANNED	LOT	EXPIRY	GIVEN
NOVA-E1 (exosome) Intra-articular or peritendinous, affected shoulder	2-4 cc total			
NOVA-M1 (MSC) Intra-articular or peritendinous, affected shoulder	per plan			

TECHNIQUE

Position: seated; posterior glenohumeral or subacromial approach

- Ultrasound guidance; confirm target (joint vs peritendinous) before injecting.
 - Aspirate before injecting; never inject into tendon substance under pressure.
 - Slow administration; sterile dressing after withdrawal.

LEFT · RIGHT — circle

SITE / SIDE

ULTRASOUND — required

GUIDANCE

22-25 G

NEEDLE USED

target 10-15 min

OBSERVATION — MINUTES

NARRATIVE — FINDINGS, ASPIRATE, DEVIATIONS FROM PROTOCOL, PATIENT RESPONSE

POST-PROCEDURE & DISPOSITION

Tolerated: WELL · WITH EVENTS — circle

Discharged: STABLE · AMBULATORY · WITH ESCORT — circle

☐ Patient note CN-PN-SHL reviewed and given at discharge

COMPLICATIONS / ADVERSE EVENTS — WRITE "NONE" IF NONE

Any adverse event also requires the adverse-event response steps in this book.

Follow-up plan. Clinical assessment at 4-6 weeks; repeat consideration at 6-12 months based on response.

FOLLOW-UP SCHEDULED FOR

PHYSICIAN SIGNATURE

PRINTED NAME & LICENSE NUMBER

DATE & END TIME

Today you received an ultrasound-guided injection of cell and exosome products in and around your shoulder. The products used are made from donated birth tissue and are not FDA-approved treatments. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

WHAT TO EXPECT

Changes are gradual — do not judge the result in the first days

FIRST 72 HOURS

Aching in the shoulder and upper arm is common and usually settles in two to three days. Sleeping on that side may be uncomfortable for a few nights.

WEEKS 1-4

Keep the shoulder moving gently — pendulum swings and easy range of motion — while avoiding heavy or overhead work.

WEEKS 4-12

Progress is assessed at your follow-up visit; strength work is added back gradually as advised.

DO

- Keep the arm gently moving; a sling is not needed
- Sleep on the other side for the first few nights
- Use cold packs 10–15 minutes at a time for comfort
- Take acetaminophen (Tylenol) for soreness if needed

AVOID

- Anti-inflammatory pain relievers for 72 hours
- Alcohol for 72 hours
- Overhead lifting and reaching for 48–72 hours
- Weight training for the shoulder until cleared

CALL US RIGHT AWAY IF

For an emergency, call 911 first

- The shoulder becomes increasingly hot, red, and swollen
- New numbness, tingling, or weakness in the arm or hand
- Fever above 101 °F or shaking chills
- Severe pain not controlled by acetaminophen and cold packs

CLINIC PHONE

AFTER-HOURS PHONE

NOTES FROM YOUR VISIT